Project Initialization and Planning Phase

Date	15 March 2024
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The Human Development Index (HDI) Predictor is a machine learning-based web application developed to predict the Human Development Index of a country using important socio-economic indicators. The system improves decision-making by providing quick predictions, graphical analysis, and an easy-to-use interface for researchers, policymakers, students, and government organizations.

Project Overview	
Objective	Develop a machine learning-based web application that predicts the Human Development Index (HDI) using Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) per Capita.
Scope	The project focuses on data preprocessing, exploratory data analysis, machine learning model training, prediction, and deployment through a Flask web application. It also provides visualizations to help users understand the factors affecting HDI
Problem Statement	
Description	Predicting a country's Human Development Index manually is difficult because it depends on multiple socio-economic indicators. Existing methods require manual analysis and interpretation of large datasets.
Impact	An automated HDI prediction system enables faster analysis, improves decision-making, supports research, and helps identify areas requiring development.
Proposed Solution	
Approach	Develop a Flask-based web application using Python and a Linear Regression model. The application accepts user inputs, processes them through the trained model, predicts the HDI score, classifies it into development categories, and displays analytical graphs.
Key Features	• Machine Learning-based HDI prediction • User-friendly Flask web interface • Data preprocessing and cleaning

	• Interactive data visualization (Heatmap, Scatter Plot, Pair Plot, Strip Plot, Distribution Plot) • HDI category classification (Very High, High, Medium, Low) • Fast and accurate prediction results

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i3 or above
Memory	RAM specifications	Minimum 4 GB RAM
Storage	Disk space for data, models, and logs	Minimum 10 GB Free Space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle
Development Environment	IDE	Visual Studio Code / Jupyter Notebook / Anaconda
Data		
Data	Source, size, format	Kaggle Human Development Index (HDI) Dataset, ~190+ records, CSV format